
A Real-Time Perception Pipeline for Adaptive Human-Robot Interaction

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Abstract

Object handover — a canonical task in human-robot interaction spanning companion, assistive, service, industrial and research co-working settings — requires the robot to adapt to user’s comfort in real time; failure to do so risks startle, avoidance, or safety incidents. Because the interaction is highly context- and person-dependent, no fixed protocol suffices, while collecting interaction data for every deployment is expensive and fine-tuning large foundation models per use case scales poorly. We present a real-time comfort-scoring pipeline that late-fuses four frozen pretrained vision models (face detection, facial expression, gaze, depth-aware pose) with only a small set of learnable parameters in the fusion layer trainable on small site-specific corpora. The fused system produces a continuous 0–100 comfort score from RGB-D video, with phase-specific fusion weights calibrated by differential evolution against a 31-recording phase-labeled dataset for on-site recalibration. The calibrated pipeline achieves highly accurate abort separation (Youden’s $J = 1.0$) on a held-out test and lifts per-recording slope-sign agreement from 17/31 to 28/31 across a full LORO sweep. We then distill the pipeline (~ 70 M parameters) into a $44\times$ -smaller single-frame RGB student that tracks the teacher at MAE 4.1 / $r = 0.80$, cross-checked against an independent vision–language-model judge that agrees with the teacher on trajectory shape ($r = 0.74$) despite having no access to the calibration setup. Together, these results offer a practical template for socially adaptive perception under data scarcity.

1 Introduction

Human-robot interaction increasingly extends beyond controlled industrial settings into companion robots for older adults, assistive care, retail and restaurant service, and human-robot collaborations in manufacturing factories and research laboratories. A common task across these settings is object handover, a representative building block for socially adaptive human-robot interaction. The challenge is that comfort during handover is highly situation- and person-dependent: the same behavior can feel natural in one moment and intrusive in another, and what one person welcomes another may reject. No fixed protocol suffices; the system must perceive each user’s comfort in real time and adapt on the fly. Otherwise, the interaction risks startling the user, eliciting avoidance behavior, or causing safety incidents.

Several factors hinder a unified end-to-end solution. First, interaction data in specific deployment is sparse and expensive: each new use case involves long continuous scenes that lack a clean framework for being framed as model-ready inputs. Second, fine-tuning a large foundation model end-to-end requires substantial parameters and computation power. Third, the traditional supervised fine-tuning requires retraining for each use case while it scales poorly. What is then needed is the decomposition of natural human behavior into a structure

perceivable by a robot in real time. We propose a three-phase framework — approach, intent signaling, service execution — where each phase inputs a different perceptual question and cues, and the entire interaction reduces to a single continuous comfort score that gates the robot’s response. Here, generalizable pipeline is integral that (i) decomposes the interaction into phases, (ii) is compatible with sparse, low-effort site-specific data, and (iii) is few-shot calibratable without retraining the underlying perception models. Late fusion of frozen pretrained unimodal models is naturally suited to this: its modularity removes the need to fine-tune any backbone per use case, and the small learnable fusion layer extracts maximum signal from the limited site-specific data available.

Three sub-problems remain. (i) **Multimodal perception** in real time: each channel has known weaknesses (facial-expression ambiguity, missing gaze when the face turns away, missed withdrawal in 2D pose) and no prior handover system [33, 35] integrates all three plus depth-based withdrawal into a single continuous comfort score on commodity RGB-D hardware. (ii) **Phase-aware weighting**: gaze is most informative before the user reaches, but during the reach gaze aversion may be a benign attention shift toward the object; a static weighting cannot resolve this. (iii) **Edge deployability**: we treat the calibrated pipeline (~ 70 M params across four backbones, GPU-tethered) as a *teacher* labeling every frame, and train a 1.58 M-param MobileNetV3-Small [30] *student* via Huber regression [32] to reproduce the score from a single RGB frame.

Contributions. (i) A deployment recipe — a phase-aware late-fusion comfort scorer over four frozen pretrained unimodal models, calibrated from a small ($n = 31$) phase-labeled RGB-D corpus by differential evolution on cached per-frame features (seconds of CPU per candidate, no backbone fine-tuning). (ii) Experimental evidence that, under our domain conditions, posture *degrades* performance, with a no-pose ablation beating its parent and a mechanism-level explanation. (iii) A teacher–student distillation in which the calibrated signal supervises a $44\times$ -smaller single-RGB-frame student (MAE 4.1 / $r = 0.80$), cross-checked against an independent GPT-5.5 VLM judge whose agreement with the teacher on trajectory shape ($r = 0.74$ per-frame) indicates the signal reflects observable behavior rather than calibration artifacts.

2 Related Work

Multimodal perception. Churamani et al. [15] fuse facial expression and speech via late fusion into an Affective Core (Grow-When-Required network) driving DDPG policy learning — the closest existing pipeline to ours, though they target dialogue and omit gaze. Surveys by Wang et al. [1], Su et al. [2], and Zhao et al. [3] systematically review 60–227 papers and identify cognition-level comfort fusion, cross-modal temporal alignment, and the perception-to-action loop as the most prominent gaps. Fusion approaches range from confidence-weighted attention [17] and bilevel NAS [18] for action recognition, through musculoskeletal IK fusion of RGB-D and IMU under occlusion [19], to vision–language models such as PaLM-E [20] that unify images, robot state, and text into a shared latent space. Our work sits in the simplest of these regimes — score-level late fusion — but treats the phase-aware fusion weights themselves as the trainable surface.

Robot-to-human handovers. The handover task has its own focused literature. Strabala et al. [33] identify a three-phase decomposition (approach, signal, transfer) that maps onto our (approach, intent, execution) phase indicator $\varphi(t)$. Cakmak et al. [34] characterize receiver preferences over object orientation, grip, and height; Ortenzi et al. [35] survey the full taxonomy of object handovers and frame receiver-side perception as an open problem distinct from grasp planning.

Comfort measurement and continuous comfort scoring. Yan and Jia [4] propose a comfort taxonomy spanning ergonomic, motion, anthropomorphism, and sociability dimensions, measured both subjectively (Likert) and objectively (HRV, EDA, EEG). Yan et al. [5] fit a multi-linear regression model from physical motion factors to predicted comfort but omit emotional signals; Lorenzini et al. [6] review ergonomic HRI in industry. Gonzalez-Santocildes et al. [7] use RL to adapt robot behavior from a continuous comfort signal, demonstrating that an online comfort score *can* drive policy. Heinisch et al. [8] release AFFECT-HRI, the

only public HRI dataset pairing physiological signals (Empatica E4 GSR/BVP) with affect labels.

Affect-aware interaction and decision-making. Spezialetti et al. [9] note that robot presence itself biases human emotional response, motivating in-situ rather than benchmark evaluation. Freire et al. [10] propose DAC-HRC, a layered adaptive control architecture with explicit personalization; Abdollahi et al. [11] show that an empathic robot (“Ryan”) improves engagement. Namba et al. [12] adopt the graduated normal/slow/stop structure we follow, but the controller is purely rule-based. For decision-making under uncertainty, Zheng et al. [13] learn POMDPs for collaboration, Pandya [14] formulates influence-aware safety as a dynamic game in the joint physical and belief space over human intention, and Brunke et al. [16] survey safe learning in robotics. Datasets for perception backbones we adopt include Gaze360 [24] and AffectNet [25]; SALSA [21], the Vernissage corpus [22], and JRDB [23] provide related social-interaction benchmarks.

How our work differs. To our knowledge, no prior system fuses facial emotion, gaze, and body posture from RGB-D video into a continuous, *phase-aware* real-time comfort score for the handover scenario, then few-shot calibrates that fusion at the deployment site, then distills the result into an edge-deployable single-modality student. Our pipeline closes this gap with score-level late fusion behind a binary abort policy, calibrated against a small ($n = 31$) site-specific recording corpus.

3 Problem Statement

Let $x_t = (I_c^{(t)}, I_d^{(t)})$ be an RGB-D frame pair captured at time t during a handover, with $I_c^{(t)}$ the color frame and $I_d^{(t)}$ the aligned depth map from an Intel RealSense D435. Each recording is annotated with up to four temporal keypoints ($t_{\text{start}}, t_{\text{signal}}, t_{\text{handover|abort}}, t_{\text{end}}$) defining a phase indicator $\varphi(t) \in \{\text{approach, intent, execution}\}$:

$$[\text{t_start}] \xrightarrow{\text{approach}} [\text{t_signal}] \xrightarrow{\text{intent}} [\text{t_handover|abort}] \xrightarrow{\text{execution}} [\text{t_end}]$$

The pipeline emits two per-frame channel scores: emotion $C_e^{(t)} = f_e(I_c^{(t)}; \Theta_E, \Theta_G)$ (facial valence/arousal from EfficientNet-B0 with frozen weights Θ_E , gaze-engagement signal from L2CS-Net with frozen weights Θ_G) and posture $C_p^{(t)} = f_p(I_c^{(t)}, I_d^{(t)})$ (MediaPipe Pose openness, depth-based withdrawal, mouth-covering). Each is independently EMA-smoothed in continuous time with $\alpha(\Delta t) = 1 - \exp(-\Delta t/\tau_{\text{ema}})$, so smoothing is FPS-independent. The integrated comfort score is the phase-aware weighted sum:

$$\tilde{C}_t = w_e^{\varphi(t)} \tilde{C}_e^{(t)} + w_p^{\varphi(t)} \tilde{C}_p^{(t)}, \quad w_e^{\varphi} + w_p^{\varphi} = 1 \quad \forall \varphi. \quad (1)$$

When detection fails on a channel, the channel decays exponentially toward a tunable target (`no_face_target`, `no_pose_target`) at a tunable rate. The robot controller collapses the policy to a binary gate at τ^* :

$$\pi(\tilde{C}_t) = \begin{cases} \text{continue} & \tilde{C}_t > \tau^* \\ \text{abort} & \text{otherwise.} \end{cases} \quad (2)$$

The learning problem is: given a small set of labeled recordings with ground-truth class $y \in \{\text{continue, abort, ambiguous}\}$, fit the 20-dimensional parameter vector θ controlling fusion weights, gaze cone, EMA τ , decay rates, and penalties so as to maximize Youden’s J on the abort decision while keeping each scenario’s comfort *trajectory* trending in the correct direction.

4 Proposed Method

4.1 System architecture

Figure 1 shows the runtime pipeline. The color frame is forked into a GPU emotion branch and a CPU posture branch, which run independently and report per-frame outputs to a shared `IntegratedComfortScorer`. The branches are deliberately decoupled so the system

```

RealSense .bag (color + aligned depth)
|
+--> GPU branch
|   +- RetinaFace -----> face bbox
|   +- EfficientNet-B0 ----> (emotion, valence, arousal)
|   +- L2CS-Net (Gaze360)-> (yaw, pitch) -> "looking-at-camera"
|
+--> CPU branch
|   +- MediaPipe Pose -----> 33 landmarks
|   +- open-posture ratio
|   +- mouth-covering check
|   +- depth-based withdrawal detector (uses I_d)
|
+--> IntegratedComfortScorer
    +- phi(t) set by JSON keypoints (calibration)
    |   or controller (deployment)
    +- continuous C(t) in [0,100]
      -> binary continue/abort at tau*

```

Figure 1: Phase-aware late-fusion architecture. Each branch operates independently; the comfort scorer combines them with phase-aware fusion weights w_e^φ, w_p^φ (the in-emotion gaze coefficient w_g^φ enters Eq. 3 separately). Phase is supplied externally — by the calibration harness during fitting and by the robot controller’s state machine during deployment.

degrades gracefully: if the face is occluded the posture channel still reports, and vice-versa. Frames in which a channel is missing trigger a per-channel exponential decay toward a tunable target rather than a hard zero, so a brief detection dropout neither flips the decision nor injects step changes into the EMA.

4.2 Per-frame instant scores

Given a detected face, EfficientNet-B0 produces a categorical emotion plus continuous valence $v \in [-1, 1]$ and arousal $a \in [-1, 1]$; L2CS-Net produces gaze (yaw, pitch) from which we derive a binary “looking-at-camera” signal $g \in \{0, 1\}$ using yaw/pitch thresholds. The instant emotion score normalized into $[0, 100]$ is

$$C_e^{(t)} = \frac{(\gamma v - \delta a + w_g^{\varphi(t)} g) + (\gamma + \delta + w_g^{\varphi(t)})}{2(\gamma + \delta + w_g^{\varphi(t)})} \cdot 100. \quad (3)$$

The phase-specific gaze coefficient w_g^φ is what makes “looking at the camera” load heavily during *intent signaling* and barely at all during *execution*, where a downward gaze toward the cup is a benign attention shift.

From MediaPipe’s 33 landmarks we compute a body-openness ratio clipped to $[0, 1]$ and check two depth-aware events: (i) mouth-covering when hand-to-mouth distance falls below $\text{face_cover_ratio} \times \text{shoulder-width}$, and (ii) sudden-withdrawal when a smoothed torso depth jumps by more than $\text{withdrawal_threshold_meters}$ over the recent history. Each fired event subtracts a fixed penalty ($\text{mouth_cover_penalty}$, $\text{withdrawal_penalty}$).

4.3 Phase-aware fusion and missing-detection decay

When a channel is missing on a given frame, $\tilde{C}_k^{(t)} \leftarrow \tilde{C}_k^{(t-1)} + (\text{target}_k - \tilde{C}_k^{(t-1)})(1 - e^{-\Delta t \cdot \text{rate}_k})$. This is the mechanism by which a walk-by recording — where face detection is sparse because the user never turns toward the camera — accumulates downward drift into the abort region without requiring positive evidence of discomfort. The four parameters ($\text{target}_e, \text{rate}_e, \text{target}_p, \text{rate}_p$) are tunable by calibration; in the deployed config the no-face decay rate is $\sim 3\times$ the hand-set default — this separates walk-by from successful handover (§6.4).

4.4 Calibration via differential evolution on cached features

The learning surface is a 20-dimensional fusion vector θ covering the phase-aware fusion weights w_e^φ, w_p^φ for $\varphi \in \{\text{intent}, \text{exec}\}$ (rescaled to sum to 1), the in-emotion gaze coefficient w_g^φ (a coefficient inside Eq. 3, *not* a fusion weight at the same level as w_e, w_p), the gaze cone, two penalty magnitudes, $\gamma, \delta, \tau_{\text{ema}}$, and the EMA / missing-detection-decay constants. Approach-phase weights are held fixed at defaults; full bounds in Appendix C.

The pipeline has four stages (split / extract / optimize / evaluate; Appendix B): **extract** runs the four perception models once per recording, caching per-frame features as parquets so that **optimize** replays from cache in pure NumPy under `scipy.optimize.differential_evolution` on the cross-validated loss below.

Loss function. For each training recording i , the runtime pipeline is replayed under parameters θ , producing per-frame integrated comfort $\tilde{C}(t; \theta)$, summarized by the mean over the recording’s intent $\cup$ execution window W_i (sidecar-defined):

$$\bar{C}_i(\theta) = \frac{1}{|W_i|} \sum_{t \in W_i} \tilde{C}(t; \theta). \quad (4)$$

The training set is split into $K = 5$ label-stratified folds. For each held-out fold f , Youden’s $J_f(\theta) = \max_{\tau \in [30, 80]} (\text{TPR}_f(\tau) + \text{TNR}_f(\tau) - 1)$ is computed on $\bar{C}_i$; the cross-validated term is $\bar{J}(\theta) = \frac{1}{K} \sum_f J_f(\theta)$. The winning variant A minimizes

$$\mathcal{L}_A(\theta) = - \left[\bar{J}(\theta) + \varepsilon \cdot \frac{\mu_{\text{sc02}}(\theta) - \mu_{\text{sc01}}(\theta)}{100} \right], \quad \varepsilon = 0.1, \quad (5)$$

pairing the classification term $\bar{J}$ with a sc02-vs-sc01 mean-margin tie-breaker. DE is gradient-free, required because $\bar{J}$ is rank-based and non-differentiable in θ ; full hyperparameters, the 20-D bound table, and the search-space derivation are in Appendix C.

Trajectory shape is an *evaluation gate* (§6.2), not part of the loss: shape-aware objective variants we tried (F/G/H/I) replaced $\bar{J}$ with tanh-encoded slope terms and all lost to variant A.

4.5 Inference

At deployment, the robot controller drives `set_phase(...)` from its own state machine (**approach** while motion-planning, **intent** while signaling, **execution** once the arm crosses the OTP plane). A 0.3s warmup after each phase transition freezes the integrated output to suppress the EMA transient. The continuous $\tilde{C}_t$ is gated against $\tau^* = 80$.

5 Experimental Methodology

Dataset. We collected 31 RGB-D recordings (`.bag`, 30FPS) on an Intel RealSense D435 mounted on a Rainbow Robotics RB-Y1 manipulator, spanning five scenarios (Table 1). Each `.bag` carries a JSON sidecar (Appendix A) with keypoints `start` / `signal` / `handover` (or `abort`) / `end` annotated post-hoc. The split is 23 train / 8 test, stratified by scenario; held-out test contains 2 sc01, 2 sc02, 1 sc03, 1 sc04, 2 sc05.

Table 1: Scenarios.

code	scenario	label	n
sc01	walk-by, no interest	abort	7
sc02	comfortable handover	continue	8
sc03	gradual discomfort during handover	ambiguous	5
sc04	sudden withdrawal mid-handover	abort	5
sc05	distracted (looking at phone)	ambiguous	6

Pretrained models. RetinaFace [27]; EfficientNet-B0 [28] / `enet_b0_8_va_mt1` (AffectNet-pretrained [25]); L2CS-Net / `L2CSNet_gaze360` (Gaze360-pretrained [24]); MediaPipe Pose [29], complexity 1. Only the 20 fusion-layer hyperparameters are learned.

Metrics. Youden’s J [31] at τ^* on the held-out test plus a τ -sweep over $[30, 80]$; per-scenario mean comfort with bootstrap 95% CIs (1000 resamples); per-recording slope sign agreement over the intentUexecution windows (correct = positive for sc02, negative otherwise); and a LORO sweep over all 31 recordings.

Baselines and ablations. `config/default.yaml` (hand-tuned midterm config, never optimized); `config/deploy.yaml` (calibrated A-no-pose winner, see §6.3); six DE objective variants A, C, F, G, H, I; a no-pose ablation pinning $w_p = 0$.

6 Results and Discussion

6.1 Calibrated config separates abort from continue cleanly

Table 2 compares the hand-tuned default with the calibrated deploy on the held-out test ($n = 8$) and full-corpus LORO ($n = 31$).

Table 2: Default vs deploy on held-out test and LORO.

metric	default	deploy	Δ
best Youden J on test	0.00 @ $\tau=30$	1.00 @ $\tau=78$	+1.00
decisions at τ^* (test)	—	TP 3 / FN 0 / FP 0 / TN 2	perfect
sc01 mean (test, abort)	69.0	50.3	−18.7
sc02 mean (test, continue)	66.6	92.8	+26.2
LORO slope-sign agreement ($n = 31$)	17/31 (55%)	28/31 (90%)	+11

The default cannot separate abort from continue at *any* $\tau \in [30, 80]$: $J = 0.0$ because the sc01 abort and sc02 continue distributions overlap and sit on the wrong side of each other (sc01 mean 69.0 > sc02 mean 66.6), so no single threshold can put aborts below and continues above it simultaneously. The calibrated deploy flips this ordering and opens a 42-point margin between sc01 and sc02 means; the same margin holds across LORO (sc01 = 50.2, sc02 = 92.7 across all 15 recordings of those two scenarios). Per-recording, under the default, sc02 *trends downward* during successful handovers (mean slope -3.05); under deploy, every test recording’s slope has the correct sign. LORO slope-sign agreement on sc02 (per-recording) jumps from 0/8 to 7/8 and on sc03 from 1/5 to 5/5.

$J = 1.0$ holds across the observed plateau $\tau \in [78, 80]$ within our $[30, 80]$ sweep range; we ship $\tau^* = 80$ as the upper edge of this plateau (most conservative continue gate, since any continue case scores well above the plateau — sc02 means are 91.6 and 94.0 on test).

6.2 Race-2 objective variants

Variant A’s objective (Eq. 5) optimizes *level* via $\bar{J}$ plus a small mean-margin tie-breaker. Variants C/F/G/H/I *replace* $\bar{J}$ (or the margin term) with shape-aware reducers built on the tanh slope term defined in §4.4; we use them to test whether trajectory direction can be optimized inside the loss without sacrificing classification. Per-scenario slope sign remains an *evaluation gate* for all variants: + for sc02, − for sc01/sc04. We raced six variants on the same 5-fold CV folds (Table 3), against seven pre-registered acceptance gates: $J \geq 0.80$, sc02 mean ≥ 80 , sc01 mean ≤ 65 , sc04 mean ≤ 65 , and per-scenario slope-sign agreement on sc02 (+), sc01 (−), sc04 (−).

Each shape-aware variant trades one scenario’s correctness for another’s (Table 3): C lifts sc02 alone by collapsing the level scale; F and G each force one scenario’s slope sign at the cost of another’s; H runs G’s objective on the 20-D space and reproduces the same failure, isolating the cause to objective *geometry* rather than search dimensionality; I corrects sc02 but lets sc01 drift positive. Variant A — level-only baseline plus mean-margin — wins 6 of 7 gates; its single miss is sc04 $\mu = 77.6$ (vs the ≤ 65 threshold), but the abort ROC at $\tau^* = 80$ is perfect and sc04’s slope (-2.82) is correctly steep, so the failed gate does not harm deployment.

The negative result: under this train budget (23 recordings, 5-fold CV), single-objective DE cannot simultaneously satisfy *level* and *shape* gates without overcorrecting on some scenario.

Table 3: Race-2 variants on held-out test. The “objective” column lists what each variant optimizes *inside* the loss; slope-sign agreement is a post-hoc gate, never minimized. Slope $\hat{\beta}$ is the linear-fit gradient of $\bar{C}(t)$ over the recording’s intent $\cup$ execution window, in comfort-points-per-second. Sign and magnitude should be read *jointly*: variant F has a strong positive sc02 slope (+2.86) but *also* a positive sc01 slope (+2.88) — every scenario rises, so the shape signal is not discriminative. Variant A has a weakly-positive sc02 (+0.10) and a clearly-negative sc01 (−0.54), which does discriminate even though the magnitudes are smaller.

var	objective	sc02 $\hat{\beta}$	sc01 $\hat{\beta}$	sc04 $\hat{\beta}$	sc02 μ	sc01 μ	sc04 μ	J	gates
A	$\bar{J} + \varepsilon \cdot \text{margin}$ (Eq. 5)	+0.10	−0.54	−2.82	89.1	55.8	77.6	1.0	6/7
C	late–early delta of $\bar{C}_i$ replaces $\bar{J}$	+2.14	−1.26	+0.47	65.8	49.5	62.7	1.0	5/7
F	$\bar{J}$ + sc02-only tanh slope	+2.86	+2.88	+1.48	73.6	53.5	51.7	1.0	4/7
G	$\bar{J}$ + mean tanh slope across scenarios	−0.75	−2.38	−3.50	81.5	75.5	77.8	1.0	4/7
H	G’s objective on the 20-dim space	−0.72	−3.35	−3.34	80.7	75.2	76.5	1.0	4/7
I	sc02-weighted ($\times 3$) tanh slope + sign penalty	+0.89	+2.61	−1.48	77.1	65.7	62.1	1.0	4/7

We promote A and expose the trade-off; the post-hoc no-pose ablation (§6.3) recovers correct sc02 trajectory direction without changing J .

6.3 No-pose ablation

Per the original plan we ran a confirmatory ablation — objective A with $w_p^{\text{intent}} = w_p^{\text{exec}} = 0$, same DE budget. The ablation clearly won (Table 4).

Table 4: A vs. A-no-pose on held-out test and LORO.

metric	A (with pose)	A-no-pose	Δ
sc02 slope (test)	+0.10	+1.05	+0.95
sc02 mean (test)	89.1	92.8	+3.7
sc01 mean (test)	55.8	50.3	better sep.
slope-sign agreement (test, per-scenario)	3/5 scenarios	5/5 scenarios	+2
sc02 LORO agreement	4/8	7/8	+3
sc01 LORO agreement	6/7	5/7	−1
J on test	1.0	1.0	—

Pose was *actively harming* the trajectory signal. DE converged to nonzero w_p in A because it marginally helped J on CV folds, but on held-out test the pose domain shift surfaced. Specifically: in cached sc02 parquets the MediaPipe withdrawal condition fired in 2–4% of intent+execution frames and mouth-cover up to 1.3% — low but non-zero where zero was expected. Each spurious fire subtracted 25 (withdrawal) or 30 (mouth-cover) from the posture sub-score, which under the 65% execution posture weight propagated directly into sc02’s integrated comfort. CV couldn’t see this because all folds share the same systematic bias. We set $w_p = 0$ but the posture module remains in the runtime as a diagnostic, and the setting can be updated when a future dataset has smaller pose domain shift.

6.4 Mechanism

Three coupled changes account for the default-to-deploy transformation:

1. **Posture weight set to 0** (see §6.3) removes the posture channel from the fusion, eliminating the MediaPipe false-fire drag on sc02.
2. **Missing-detection decay rate $\sim 3\times$ faster.** The default’s ~ 6.7 s time constant left sc01 (with $\sim 40\%$ face-detection rate over the intent $\cup$ execution window) drifting only slowly toward the no-face target. Deploy pulls comfort toward `no_face_target` ≈ 32.5 aggressively whenever the face is absent, so sc01’s missing-face frames accumulate a substantial downward drag (LORO mean drops from 69.2 to 50.2); sc02’s 68–72% face-detection rate leaves the boosted rate mostly idle (LORO mean 92.7). This is the

largest single contributor to the 42-point separation, and what justified promoting the four hardcoded decay constants to tunables.

3. **Wider gaze cone and rebalanced emotion weights:** intent emotion $0.6 \rightarrow 0.882$; intent gaze coefficient $0.6 \rightarrow 0.808$; yaw cone $25^\circ \rightarrow 31.7^\circ$; pitch cone $20^\circ \rightarrow 29.9^\circ$. With pose removed, the widened cone matters because sc02 users do not always fixate directly on the camera while reaching.

Smaller supporting changes: `face_cover_ratio` $0.6 \rightarrow 0.85$, `withdrawal_penalty` $25 \rightarrow 11.9$, γ $0.5 \rightarrow 0.11$, δ $0.3 \rightarrow 0.006$, τ_{ema} $0.5 \rightarrow 0.286$.

Taken together, these three mechanism-level changes show that a 31-recording phase-labeled site corpus, combined with frozen pretrained unimodal backbones and a low-dimensional learned fusion layer, is sufficient to convert generic perception models into a deployment-specific comfort scorer, without backbone fine-tuning and a large training set.

7 Edge Deployment

The calibrated pipeline works, but it is not where we want to ship: ~ 70 M parameters across four pretrained vision models, sustaining 30 FPS requires an RTX-class GPU laptop tethered to the robot. We therefore distil the calibrated signal into an edge-deployable single-RGB-frame student. Because student $\leftrightarrow$ teacher agreement only confirms that the student reproduces the teacher — not that the teacher is right — we additionally probe the teacher’s labels with an independent vision–language-model (VLM) judge. The VLM does *not* supervise the student; it is used only post-hoc as a cross-check on trajectory shape. The full pipeline is in `V3_PIPELINE.md`; empirical numbers are in `V3_TEST_REPORT.md`; 10 side-by-side comparison videos and the verbatim VLM system prompt are mirrored on the HuggingFace dataset `yirantao1000/mmai-comfort-handover-v3` [26].

7.1 Teacher and student

Teacher. `config/deploy.yaml` run over every recording, emitting one dense ~ 30 Hz per-frame $\hat{C}_t \in [0, 100]$ per RGB frame. Labels embed the full multimodal context (phase-aware fusion, EMA, missing-detection decay) even though only the RGB frame is exposed to the student.

Student. MobileNetV3-Small with a single-scalar regression head, 1.58 M parameters, ingesting one RGB frame and emitting a 0–100 comfort score — no depth, no temporal context, no phase signal. Headline parameter reduction: $70 \text{ M} / 1.58 \text{ M} \approx 44\times$ (we report parameter count only; FLOPs and on-robot inference time are not measured).

The distillation corpus merges the 31 calibration recordings with 26 additional bags collected in a second session (2026-05) for a total of 57 bags. `splits/v2.json` (`scripts/make_v2_split.py`, seed 42) draws 47 train / 10 test from this pool, with 2 test bags per scenario drawn deterministically.

Huber loss against the teacher’s per-frame score, 10 epochs, frame subsampling at 5 FPS, JPEG-cached frames preprocessed via standard MobileNet pipeline.

7.2 VLM as independent cross-check

To probe whether the teacher’s per-frame labels reflect observable behavior in the frames rather than calibration setup, we run an independent VLM (GPT-5.5 via the Responses API) over the same recordings, prompted with raw frames plus timestamps. Four annotator design tricks: (i) sliding $K = 6$ window with stride 3, so each frame is averaged across two windows; (ii) ordinal soft distribution $p = [p_1, \dots, p_5]$ summing to 1, preserving uncertainty; (iii) per-window natural-language rationale (one sentence per call), enabling spot-checks; (iv) causal EMA smoothing ($\tau = 0.6$ s) producing a real-time-style score at ~ 2 Hz.

The VLM’s output is never used to train the student. Its sole role is to provide an information-restricted second opinion on per-frame comfort trajectory, which we use to assess whether the teacher’s signal generalizes beyond the specifics of our calibration setup.

7.3 Results

Table 5 reports agreement between teacher (T), VLM (V), and student (S) on the 10 held-out test bags. We report MAE only for T $\leftrightarrow$ S, because the student is trained to reproduce the teacher’s 0–100 scale and absolute deviation is the relevant quantity. The VLM is an independent estimator with no alignment to the teacher’s calibration, so its absolute level is not a meaningful comparison target; we report only Pearson r for pairs involving V, which captures trajectory-shape agreement.

Table 5: Three-way agreement on the 10 held-out test bags. MAE on 0–100 reported only for T $\leftrightarrow$ S; Pearson r is the per-frame trajectory correlation.

split	n	T $\leftrightarrow$ V (r)	S $\leftrightarrow$ T (MAE / r)	S $\leftrightarrow$ V (r)
all test bags	10	0.74	4.1 / 0.80	0.75

(1) **The student tracks the teacher closely (S $\leftrightarrow$ T MAE 4.1 / r 0.80).** One frame, no depth, no phase, 44 $\times$ fewer parameters, and the per-frame trajectory still follows the multimodal teacher. This is an *in-distribution* fit-quality measure by construction — the student was trained on T’s labels via Huber regression — so $r = 0.80$ is not directly comparable to the $r = 0.74$ between two *independent* estimators below.

(2) **Teacher and VLM agree on trajectory shape ($r = 0.74$).** An independent estimator with no information sharing reaches the same per-frame trajectory direction as the teacher, providing evidence that the teacher’s signal reflects observable behavior in the frames rather than calibration artifacts.

7.4 Scope of the distillation claim

The student cannot see depth, history, or phase, which accounts for the residual MAE of 4.1. This section establishes a labeling and validation *recipe*; on-robot latency, power, and closed-loop controller behavior are left as future work (§8). Distillation also does not validate the teacher itself — a wrong calibration is faithfully reproduced by the student.

8 Conclusion and Future Directions

Takeaways. Few-shot calibration via cached-feature DE lifts Youden J from 0.00 to 1.0 on held-out test and LORO slope-sign agreement from 17/31 to 28/31, validating phase-aware late fusion over frozen pretrained backbones as a deployment recipe. The calibrated teacher then supervises a 44 $\times$ -smaller MobileNetV3 student (MAE 4.1 / $r = 0.80$), with an independent GPT-5.5 VLM judge agreeing on trajectory shape ($r = 0.74$). Three lessons: (i) dataset design and phase-aware labeling matter more than dataset size when calibrating fusion; (ii) frozen unimodal models plus a learned low-dimensional fusion layer suffice to extend perception to a new functional definition; (iii) an optimized late-fusion model is a credible supervisor for an edge-deployable student.

Limitations. Small corpus ($n = 31$ calibration, $n = 8$ held-out; 47/10 for distillation), five scenarios, one room, one camera mount. The level/shape trade in §6.2 reflects our train budget more than the method. The distillation inherits the teacher’s calibration: if $w_p = 0$ is wrong for a future site, the student reproduces the error. Cross-session generalization is untested — only sc01 spans both 2026-04 and 2026-05 sessions, partially confounding session and scenario. On-robot latency/power/controller behavior of the student is unmeasured.

Future directions. (1) Recover pose via hysteresis on withdrawal/mouth-cover triggers and site-specific tuning, productized as a one-command per-deployment recalibration routine. (2) Close the perception-to-action loop by wiring `live_comfort.py` into the controller, and restore a three-tier continue/slow/abort policy [12] via a second threshold τ_{slow} . (3) Double the corpus to ~ 60 recordings, splitting sc03/sc05 into named-class targets. (4) Deploy the §7 student on onboard compute and re-run the live comparison.

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A Sidecar JSON format

Each `.bag` recording is paired with a JSON sidecar carrying the keypoints used by the calibration harness to set $\varphi(t)$. For `sc02–sc05` (full handover or attempted handover):

```
{
  "scenario_code": "SC-02",
  "scenario_name": "Comfortable handoff",
  "duration_seconds": 12.5,
  "realsense_serial": "838212070352",
  "labels": {
    "start_time": 3.04,
    "signal_time": 4.14,
    "handover_time": 11.41,
    "end_time": 15.88
  }
}
```

For `sc01` (walk-by, no handover) the structure is `(start, signal, abort)` with `handover_time` absent. The labeling tool overwrites `labels` post-hoc as the annotator steps through the video and presses S/I/H/E (or A) hotkeys.

B Calibration command sequence

```
python scripts/calibrate.py split
# 75/25 stratified split -> data/split.json
python scripts/calibrate.py extract
# GPU pass; caches per-frame features -> cache/*.parquet
python scripts/calibrate.py optimize
# CPU-only DE on cached features -> config/deploy.yaml
python scripts/calibrate.py evaluate
# held-out test -> reports/calibration_report.json
```

Once `extract` has run, `optimize` and `evaluate` iterate at CPU speeds. Full DE budget (50 iter $\times$ pop 15) for one objective variant takes ~ 20 min single-threaded and trivially parallelizes across variants.

C 20-dimensional calibration vector and bounds

```
theta = (w_e^intent, w_p^intent, w_g^intent,
         w_e^exec, w_p^exec, w_g^exec,
         yaw_thresh, pitch_thresh, face_cover_ratio,
         mouth_cover_penalty,
         withdrawal_thresh_m, withdrawal_penalty,
         gamma, delta, tau_ema, posture_drop_gate,
         no_face_target, no_face_rate,
         no_pose_target, no_pose_rate)
```

Representative bounds: `no_face_rate` $\in [0.05, 0.50]$; `yaw_thresh` $\in [5^\circ, 40^\circ]$; `pitch_thresh` $\in [5^\circ, 40^\circ]$; `face_cover_ratio` $\in [0.3, 0.9]$. Each $w_e^\varphi + w_p^\varphi = 1$ is enforced by L1 rescaling of the DE candidate at evaluation time.

D Calibration process documentation in the repository

The main paper compresses a two-race calibration sequence into Tables 3 and 4. The unfiltered working notes are preserved in the repository for any reader who wants the full decision trail:

- `reports/20260419_1124_calibration_process_braindump.md` — the unedited working narrative: starting point, why $J = 1.0$ wasn't the end of the story, the Race 1 trade between the gate-count winner ($J = 1.0$ but slightly-negative sc02 slope, -0.10) and the shape-aware finalist (clearly-positive sc02 slope, $+0.93$, but $J = 0.667$), how the hardcoded drag-ceiling problem was diagnosed between races, the 15 $\rightarrow$ 20-dim expansion, Race 2 per-variant analysis, the surprise no-pose ablation, default-vs-deploy mechanism, process failures (`Path.relative_to` crash, default report-paths dumping to project root, the 02:30-vs-09:56 wake gap), and an honest list of claims we still wouldn't make.
- `reports/calibration_comparison.tex` (compiled to `calibration_comparison.pdf`) — the formal calibration-impact writeup: the 31-recording LORO before/after table reproduced from Table 2, the 20-dimensional search-space table grouped by role (fusion weights, sub-score coefficients, gaze cone, posture event detectors, penalties, EMA, missing-detection decay), and the loss derivation (Eq. 5) with a variable-summary table mapping every symbol to its meaning. Authoritative source for the objective.
- `reports/RUNS.md` — the calibration-run index: one row per (config, held-out test metrics) pair across both races, with the seven-gate ranking table, the LORO breakdown for A and A-no-pose, and the archive chain from `config/deploy.baseline.yaml` $\rightarrow$ `config/deploy.prerace.yaml` $\rightarrow$ `config/deploy.A_withpose.yaml` $\rightarrow$ `config/deploy.yaml`.

- `reports/default_vs_deploy.md` — the head-to-head writeup of `default.yaml` vs. the calibrated `deploy.yaml`: full parameter-by-parameter table of what changed, the three-mechanism explanation, per-scenario level and slope deltas on both test and LORO, and the bottom-line that the default config produces the *wrong ordering* and the *wrong trajectory direction*, not merely a suboptimal threshold.

These documents are part of the repository deliverable per the course’s GitHub-repository requirement, not appendix figures of this paper; they are referenced here so reviewers can audit any specific design decision against the working record without recovering it from `git log`.

E Repository structure

```
src/
  pipeline.py      # IntegratedPipeline orchestrating branches
  comfort.py       # IntegratedComfortScorer
                  # (phase-aware fusion + EMA + decay)
  phases.py        # PHASES, PhaseWindows, sidecar JSON parsing
  detectors/       # face / emotion / gaze / pose adapters
  visualization.py # HUD overlay
  bag_source.py    # RealSense .bag playback abstraction
scripts/
  calibrate.py      # entry point for the four-stage pipeline
  extract_features.py # Stage A: GPU pass, caches parquets
  optimize_params.py # Stage B: differential evolution
  evaluate_test.py  # Stage C: held-out test
  loro_eval.py      # full-corpus LORO sweep
  race_objectives.py # parallel race over objective variants
  run_bag.py        # interactive playback of a recording
  live_comfort.py   # reference live deployment
config/
  default.yaml      # hand-tuned baseline (never calibrated)
  deploy.yaml       # calibrated A-no-pose winner (deployed)
  deploy.{A_withpose, prerace, baseline,
            A,B,C,F,G,H,I}.yaml # archived variants
reports/
  default_vs_deploy.md # head-to-head writeup
  RUNS.md              # index of every run
  20260419_1124_calibration_process_braindump.md
  eval_*.json, loro_*.json,
  *_race*.yaml,json}  # raw outputs
```